HAROL MAXILAN

Data Science Undergraduate | Data Engineering & AI/ML Engineer

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PROFILE

Data Science undergraduate at Sabaragamuwa University of Sri Lanka with deep hands-on expertise spanning Data Engineering, Machine Learning, Natural Language Processing, and Autonomous AI Systems. Proven experience designing Medallion Architecture (Bronze–Silver–Gold) lakehouse ETL pipelines, automating data quality auditing frameworks, fine-tuning transformer models (BERT, DistilBERT), and developing real-time computer vision interfaces. Proficient in architecting Level 3 multi-agent autonomous workflows using LangGraph and LLMs, alongside high-throughput FastAPI microservices and scalable relational database management in MySQL. Seeking internship opportunities in Data Engineering, AI/ML Engineering, or Data Science.

EDUCATION

BSc (Hons) in Data Science — Sabaragamuwa University of Sri Lanka, Faculty of Computing *2024 – 2028 (Expected)*

Relevant Coursework: Database Systems, Data Warehousing, Data Science, Statistics, Computer Programming, Machine Learning, Artificial Intelligence

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, JavaScript, PHP

Data Engineering & Big Data: ETL/ELT Pipelines, Medallion Architecture (Bronze–Silver–Gold), Apache Spark, PySpark, Batch Processing, Data Quality Auditing, Referential Integrity, Deduplication, Data Transformation, Parquet, Star/Snowflake Schema

Machine Learning & Deep Learning: Scikit-learn, PyTorch, Classification, Regression, Feature Engineering, Cross-Validation, Hyperparameter Tuning, Model Evaluation, SMOTE, XGBoost, AutoML, MLflow

Generative AI & LLMs: LangChain, LangGraph, Multi-Agent AI Systems, RAG (Retrieval-Augmented Generation), Prompt Engineering, Pydantic, Vector Search (FAISS, ChromaDB), Google Generative AI

Natural Language Processing (NLP): Hugging Face Transformers, BERT, DistilBERT, TF-IDF, Text Classification, Tokenization, Class Imbalance Handling

Computer Vision: OpenCV, MediaPipe, Hand Gesture Recognition, Facial Emotion Detection, CNNs, Real-Time Inference

Databases & Data Analysis: MySQL, PostgreSQL, MariaDB, Pandas, NumPy, Matplotlib, Plotly, Exploratory Data Analysis (EDA), Statistical Modeling

Cloud, DevOps & Tools: AWS (S3, EC2), FastAPI, Streamlit, Docker, Git, GitHub, Linux, Jupyter Notebook, VS Code, CI/CD, REST APIs, Pytest

PROJECTS

NLP-Based Emotion Detection Pipeline *2025*

Tech: *Python, Scikit-learn, Hugging Face Transformers, BERT, DistilBERT, TF-IDF, PyTorch* | GitHub Repository

- Engineered an end-to-end NLP classification pipeline benchmarking a TF-IDF + Logistic Regression baseline (0.76 F1) against fine-tuned BERT and DistilBERT transformer models.
- Mitigated severe class imbalance across 20,000+ text samples using class-weighted loss functions, custom WeightedTrainer, and label smoothing, boosting minority-class Recall by 24%.
- Achieved a 0.88 Macro F1-score and 91.2% Weighted F1 on held-out test data, outperforming baseline models by 15.8% in multi-class emotional intent classification.

Image-Based Emotion Detection *2025*

Tech: *Python, OpenCV, Deep Learning, CNN, Streamlit, h5py* | GitHub Repository

- Engineered a real-time facial emotion recognition pipeline using OpenCV Haar Cascades for facial detection, bounding-box localization, and frame-level preprocessing.
- Trained and optimized a deep Convolutional Neural Network (CNN) across 35,000+ facial images (FER-2013), achieving 87.4% validation accuracy across 7 discrete emotional classes.
- Built an interactive Streamlit application supporting both batch image analysis and live webcam video streaming with <25ms real-time inference latency and dynamic confidence scores.

Friday — Automated Machine Learning Pipeline *2025*

Tech: *Python, Scikit-learn, FastAPI, Google Generative AI, Pandas, NumPy* | GitHub Repository

- Architected an automated machine learning (AutoML) pipeline with a high-throughput FastAPI backend for autonomous data ingestion, dynamic feature encoding, and model selection.
- Built automated preprocessing and feature engineering pipelines in Pandas and Scikit-learn across 15+ benchmark datasets, cutting manual model benchmarking time by 70%.
- Integrated Google Generative AI (Gemini API) to automatically synthesize diagnostic analytical reports and performance interpretations for stakeholders in <3 seconds.

BEE — Hand Gesture-Controlled System Interface *2025*

Tech: *Python, MediaPipe, OpenCV, PyAutoGUI, Real-Time Computer Vision, HCI* | GitHub Repository

- Built a touchless human-computer interaction (HCI) interface using MediaPipe Hands and OpenCV, tracking 21 3D hand landmarks from live webcam streams at 30+ FPS with <20ms latency.
- Designed a geometric coordinate classifier achieving 96% recognition accuracy across 7 discrete multi-finger gestures (browser navigation, tab switching, media playback, sleep mode).
- Engineered an asynchronous action dispatch engine using PyAutoGUI and system-level APIs for cross-platform (Windows, macOS, Linux) desktop automation and shortcut execution.

CareerCompass AI — Autonomous Career Coaching System *Aug 2026*

Tech: *Python, LangGraph, LangChain, Multi-Agent AI, FastAPI, React, LLMs, Pydantic* | GitHub Repository

- Architected a Level 3 autonomous multi-agent career coaching ecosystem orchestrated via LangGraph state graphs with 5 collaborative agents (CV analyzer, gap assessor, interviewer, scraper, roadmap generator).
- Implemented real-time live job and course retrieval using Tavily web search APIs and dynamic resume parsing with Pydantic schema validation.
- Designed deterministic JSON validation loops with self-correcting prompt mechanisms, achieving a 100% structured schema compliance rate across 50+ simulated mock interview sessions.

Syndicate DataStorm — Lakehouse Pipeline & Demand Prediction Model *2025*

Tech: *Python, Medallion Architecture, Pandas, Scikit-learn, OpenStreetMap API, ETL* | GitHub Repository

- Designed and deployed an end-to-end Lakehouse ETL pipeline using Medallion Architecture (Bronze–Silver–Gold) processing 2.4M+ transactional records across 20,000+ retail outlets.
- Engineered automated data quality gates enforcing referential integrity, deduplication, and schema validation, quarantining ~4.8% invalid records into Silver dead-letter tables.
- Extracted geospatial Point-of-Interest (POI) data via OpenStreetMap Overpass API, generating spatial density features that improved downstream demand forecasting RMSE by 18%.

NexLab — Smart University Resource Allocation System *2025 – 2026*

Tech: *PHP, MySQL, JavaScript, HTML5/CSS3, Priority Scoring Engine, AI Intelligence Dashboard* | GitHub Repository

- Led full-stack architecture as Team Leader (Team Predictra, CIPHER 2.0 Finalist) for an enterprise university lab resource allocation platform serving 500+ student/faculty requests.
- Engineered a multi-factor priority scoring engine with automated conflict resolution, round-robin slot allocation, and waitlist auto-promotion, eliminating 100% of double-booking conflicts across 4 user roles.
- Built an administrative Intelligence Dashboard delivering 7-day predictive demand forecasting, utilization heatmaps, and behavioral anomaly detection using dynamic threshold filters.

F1 Super Intelligence — Data Analytics & Precomputation Pipeline *2025*

Tech: *Python, Pandas, Plotly, Pytest, Batch Processing, Data Quality Auditing* | GitHub Repository

- Constructed a high-performance batch ETL pipeline aggregating and precomputing 1.2M+ Formula 1 telemetry data points across 24 Grand Prix rounds for sub-second dashboard rendering.
- Developed a 56-test Pytest data auditing suite verifying sensor completeness, race strategies, and lap timing anomalies, achieving 100% telemetry integrity coverage.
- Built pace degradation and tire telemetry models rendering interactive Plotly dashboards, reducing analytical query response latency from 4.2s to <350ms.

AWARDS & ACHIEVEMENTS

- Nexora Ideathon — 3rd Place | Sabaragamuwa University of Sri Lanka (2025)
- CIPHER 2.0 Hackathon & Case Study — Finalist | Informatics Institute of Technology (IIT) (2025)
- J'PuraXtreme 3.0 Hackathon — Finalist | University of Sri Jayewardenepura (2025)
- CodeRally 7.0 Hackathon — Intermediate Tier Finalist | Informatics Institute of Technology (IIT) (2025)
- Code Arena Hackathon — Finalist | General Sir John Kotelawala Defence University (KDU) (2025)

LEADERSHIP & VOLUNTEER EXPERIENCE

- AWS Logistics and Operations Lead | Sabaragamuwa University of Sri Lanka (Oct 2026 – Present)
- Club Cricket Team Captain | VSC, Vavuniya (Oct 2025)
- Clash of Pawns Chess Competition Mentor | AUSDAV, Vavuniya (Jun 2025)
- Zone C University Ambassador — IEEE INSL | Sabaragamuwa University of Sri Lanka (Apr 2025)
- Pentathlon 2.0 Logistics Lead | AUSDAV, Vavuniya (Dec 2023)

REFERENCES

Ms. T. Kayalvizhy — Lecturer, Sabaragamuwa University of Sri Lanka | kayalvizhy@foc.sab.ac.lk | +94-754-376-923

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